# Industrial Vision

## Lecture 5 - Harris Corner Detector II

Eigen-analysis, corner response and sub-pixel refinement

+ board

# Where we are

In the last lecture we asked what makes a patch trackable. A window is shifted by $(\Delta x, \Delta y)$ and the change in its content is measured by the sum of squared differences:

$$C(\Delta x, \Delta y) = \sum_{(x,y)\in W} \big[ I(x+\Delta x,\, y+\Delta y) - I(x,y) \big]^2$$

A first-order Taylor expansion, $I(x+\Delta x, y+\Delta y) \approx I + \Delta x\, I_x + \Delta y\, I_y$, converted this into a quadratic form:

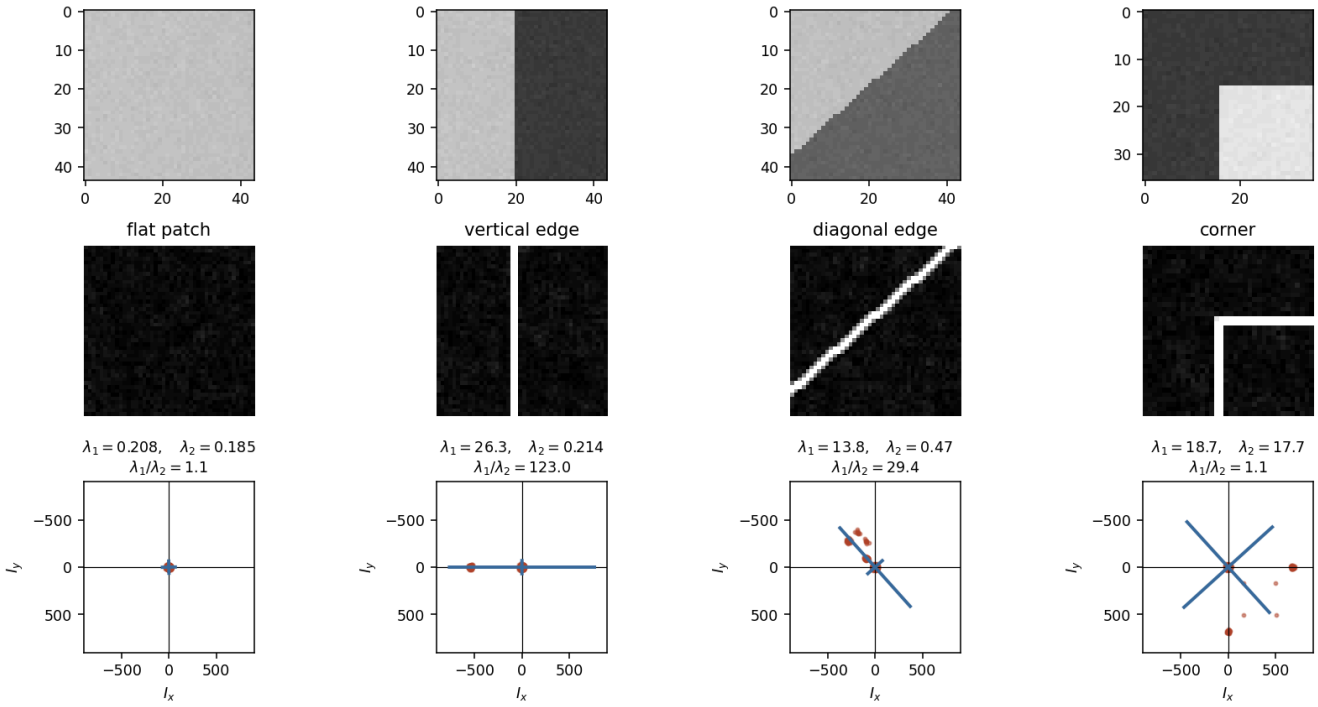
$$C(\Delta x, \Delta y) \;\approx\; \begin{bmatrix} \Delta x & \Delta y \end{bmatrix} \underbrace{\begin{bmatrix} \sum I_x^2 & \sum I_x I_y \\ \sum I_x I_y & \sum I_y^2 \end{bmatrix}}_{R \text{ (structure / gradient matrix)}} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix}$$

Whether the patch is flat, an edge or a corner is encoded entirely in this $2 \times 2$ matrix built from the gradients. **Today we interpret $R$.** Its eigenvalues are combined into a single response value, from which corners are detected and then refined to sub-pixel accuracy.

+ board

# The distribution of the gradients

For a given patch, we plot the values $(I_x, I_y)$ of every pixel as a scatter diagram.




The plots are computed from real patches of the test scene. The blue lines show the eigenvector directions of R. Above each panel are the eigenvalues $\lambda_1 \geq \lambda_2$ of R for that patch, in units of $10^6$ (raw sums over the window, so their absolute size depends on the window and on the gradient kernel; only their relative size matters).

# The eigen-decomposition $R = P \Lambda P^\top$

Since $R$ is symmetric it admits an orthogonal eigen-decomposition, with $\Lambda = \mathrm{diag}(\lambda_1, \lambda_2)$, where $\lambda_1 \geq \lambda_2 \geq 0$, and with the eigenvectors as the columns of $P$.

- The eigenvector $p_1$, belonging to $\lambda_1$, gives the direction of maximum gradient spread. For an edge this direction is **orthogonal to the edge**.
- The eigenvector $p_2$ gives the orthogonal direction. For an edge it lies along the edge, where there is very little variance.

| Patch | $\lambda_1$ | $\lambda_2$ |
|---|---|---|
| Flat region | small | small |
| Edge, in any orientation | **large** | small |
| Corner | **large** | **large** |

Consider the panel showing the diagonal edge once more. Rotating the edge rotates the eigenvectors but leaves the pattern of eigenvalues unchanged. The eigenvalues therefore constitute an orientation-free summary of the patch.

+ board

# The corner response function

A single quantity separates all three cases (Harris and Stephens, 1988):

$$M \;=\; \lambda_1\lambda_2 \;-\; \kappa\,(\lambda_1+\lambda_2)^2, \qquad \kappa \approx 0.04 \text{ to } 0.06$$

- On a **flat region** both $\lambda_1$ and $\lambda_2$ are small, so both terms are negligible and $M \approx 0$.
- On an **edge** $\lambda_1$ is large and $\lambda_2 \approx 0$, so the product $\lambda_1\lambda_2$ vanishes while the term $\kappa\lambda_1^2$ remains, and $M$ is significantly negative.
- On a **corner** both eigenvalues are large, so the product dominates the term damped by $\kappa$, and $M$ is significantly positive.

The sign of $M$ therefore classifies the patch as flat, edge or corner, and its magnitude gives the **corner strength**. The parameter $\kappa$ is set empirically. A larger value of $\kappa$ requires the pair $(\lambda_1, \lambda_2)$ to be more nearly equal before a patch is declared a corner.

+ board

# The response evaluated on the four patches

Take the eigenvalues measured earlier from the four real patches and substitute them, with $\kappa = 0.05$. Eigenvalues are in units of $10^6$, so $M$ is in units of $10^{12}$.

| Patch | $\lambda_1$ | $\lambda_2$ | $\lambda_1\lambda_2$ | $\kappa(\lambda_1+\lambda_2)^2$ | $M$ | Verdict |
|---|---|---|---|---|---|---|
| Flat region | 0.21 | 0.19 | 0.04 | 0.01 | $+0.03$ | flat |
| Vertical edge | 26.27 | 0.21 | 5.61 | 35.07 | $-29.5$ | edge |
| Diagonal edge | 8.83 | 0.35 | 3.11 | 4.22 | $-1.1$ | edge |
| Corner | 18.70 | 17.75 | 331.8 | 66.4 | $+265.4$ | **corner** |

Observations:

- On the flat patch $M$ is **positive**, not zero, since the two small eigenvalues are nearly equal and the product just exceeds the penalty. Its value is nevertheless 0.01% of the corner response. The sign alone is therefore not sufficient, and a **threshold** is required.
- On the edges the penalty term overwhelms the product, and the more elongated the distribution the more negative the value. Note that the vertical edge, being the stronger of the two, is the more negative.

+ board

# The response requires no eigen-decomposition

Computing $\lambda_1$ and $\lambda_2$ at every pixel would be wasteful. It is unnecessary, because

$$\lambda_1 \lambda_2 = \det R \qquad\qquad \lambda_1 + \lambda_2 = \operatorname{tr} R$$

$$\Rightarrow \quad M = \det R - \kappa \left(\operatorname{tr} R\right)^2 = \left(\sum I_x^2\right)\left(\sum I_y^2\right) - \left(\sum I_x I_y\right)^2 - \kappa\left(\sum I_x^2 + \sum I_y^2\right)^2$$

This requires only four products and a few sums per pixel. The value of the combination of determinant and trace is that the eigenvalues explain what the response measures, while the arithmetic never requires the eigenvectors to be computed.

+ board

# From response map to corners

**Step 1: thresholding**

Points whose corner strength exceeds $T$ are retained. Without a threshold every point with $M > 0$ would be reported as a corner. The threshold is set *relative* to the largest response in the image, typically at 1%.

**Step 2: non-maximum suppression**

A $3 \times 3$ neighborhood is examined around each candidate, and the candidate is retained only if it is the **local maximum**.

**Why is it required?** Around a true corner an entire cluster of pixels exceeds the threshold, the same difficulty encountered with edge thresholds. We require **one response per corner**.

**Reading a detector's output.** The square often drawn at a corner has area proportional to its strength: the corner is the **center point**, and the square makes no claim about the region.

```
HARRIS(image I, kappa, window w, rel. threshold t)

 # 1. gradients                            (Lecture 3)
 Ix <- Sobel_x * I ;  Iy <- Sobel_y * I

 # 2. structure matrix R, one per pixel  (Lecture 4)
 Sxx <- windowsum_w(Ix.Ix)               # weighted sum
 Syy <- windowsum_w(Iy.Iy)               # over the window
 Sxy <- windowsum_w(Ix.Iy)               # w at every pixel

 # 3. response map -- no eigen-decomposition
 det <- Sxx.Syy - Sxy^2                  # = L1.L2
 tr  <- Sxx + Syy                        # = L1+L2
 M   <- det - kappa * tr^2

 # 4. threshold, then 3x3 non-max suppression
 T <- t * max(M)                         # t = 0.01
 C <- { p : M(p) > T and M(p) = max M(3x3 around p) }

 # 5. sub-pixel refinement                (next slides)
 for p in C:  p <- p - H^-1 grad_M  at p

 return C

 ## Every line is a whole-image array operation.
```

The window $w$ is the 2nd Gaussian of the previous lecture. Most implementations, `cv2.cornerHarris` among them, substitute a uniform box, which is cheaper.

+ board

# The need for sub-pixel localization

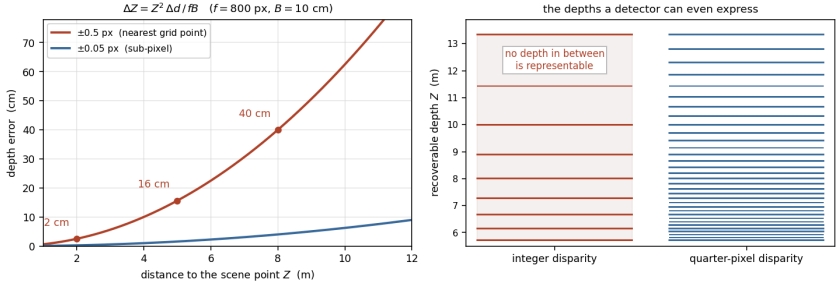
Non-maximum suppression has given us a single response per corner, but the position it reports is an **integer**. The response map lives on the pixel grid, whereas the maximum of the underlying continuous surface almost never falls on a grid point. Rounding to the nearest sample costs up to **half a pixel** in each coordinate.

That half pixel is not an aesthetic complaint. In the geometric problems - stereo, calibration, panoramas - corner positions are the *inputs* to linear systems, and the geometry amplifies the error.

Depth follows from disparity, $Z = fB/d$, so

$$\Delta Z = \frac{Z^2}{fB}\,\Delta d$$

This grows with the **square of the range**. And an integer-only detector cannot even express a depth lying between two neighboring planes.

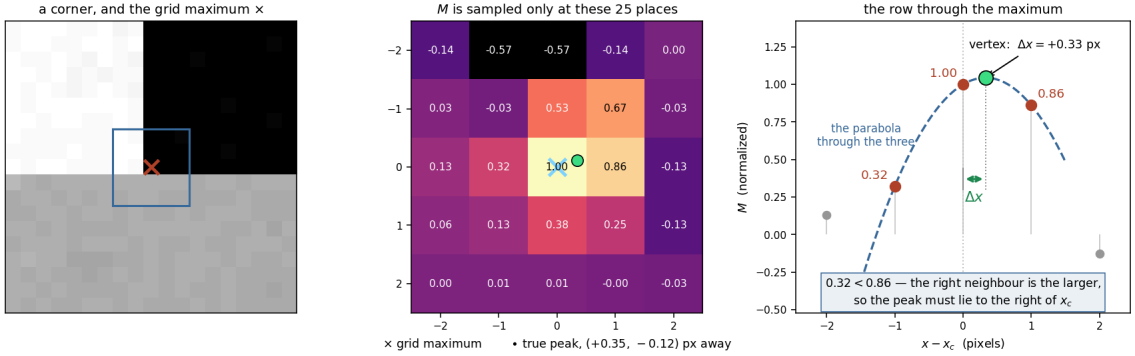



A modest stereo rig, f = 800 px and B = 10 cm. Left: the depth error produced by a localization error of half a pixel, against the sub-pixel case. Right: the depths that each can represent at all.

+ board

# We measure only on the grid — yet the samples locate the peak

$M$ can be evaluated only at integer $(x, y)$. Where is a fractional answer supposed to come from?




The sub-pixel information is carried by the **values** of the samples, not by their positions. A peak sitting off center leaves its two neighbors *unequal*, and the size of that imbalance says how far off center it lies. The parabola through the three samples has its vertex at

$$\Delta x = \frac{1}{2} \cdot \frac{f(x_c - 1) - f(x_c + 1)}{f(x_c - 1) - 2f(x_c) + f(x_c + 1)} = -\frac{f_x}{f_{xx}}$$

Only ratios of differences enter, so the scale of $M$ is irrelevant. It remains to do this in two dimensions, where the coordinates are coupled.

# Taylor expansion of the response surface

Let the maximum found on the grid be $(x_c, y_c)$ and the true maximum be $(x^*, y^*) = (x_c + \Delta x,\ y_c + \Delta y)$, with $|\Delta x|, |\Delta y| \lesssim 1$. Regarding the response as a continuous function $f(x, y)$ and expanding about $(x_c, y_c)$ to second order gives

$$f(x^*, y^*) \ = \ f(x_c, y_c) + \Delta x\, f_x + \Delta y\, f_y + \cdots$$

All derivatives here are evaluated at $(x_c, y_c)$.

Since $(x^*, y^*)$ is a **maximum**, the gradient of $f$ vanishes there. Differentiating the expansion with respect to $\Delta x$ and $\Delta y$ gives

$$\begin{aligned} 0 = f_x(x^*, y^*) \ &\approx \ f_x(x_c, y_c) + \Delta x\, f_{xx}(x_c, y_c) + \Delta y\, f_{xy}(x_c, y_c) \\ 0 = f_y(x^*, y^*) \ &\approx \ f_y(x_c, y_c) + \Delta x\, f_{yx}(x_c, y_c) + \Delta y\, f_{yy}(x_c, y_c) \end{aligned}$$

These are two linear equations in the two unknowns $\Delta x$ and $\Delta y$, and every other quantity can be measured from the response map.

+ board

# Solution of the $2 \times 2$ system

$$\begin{bmatrix} f_{xx}(x_c, y_c) & f_{xy}(x_c, y_c) \\ f_{yx}(x_c, y_c) & f_{yy}(x_c, y_c) \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} = - \begin{bmatrix} f_x(x_c, y_c) \\ f_y(x_c, y_c) \end{bmatrix} \quad \Rightarrow \quad \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} = -H^{-1} \nabla f$$

In practice $f_{xy} = f_{yx}$. The refined corner position is $(x_c + \Delta x,\ y_c + \Delta y)$.

The derivatives of the response map are computed discretely, using 1D kernels applied along the rows and columns:

$$f_x : \tfrac{1}{2}[1 \ \ 0 \ \ -1] \qquad f_{xx} : [1 \ \ -2 \ \ 1] \qquad f_{yy} : [1 \ \ -2 \ \ 1]^\top \qquad f_{xy} : \tfrac{1}{4} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

The kernel $[\,1 \ -2 \ \ 1\,]$ is the 1D second difference obtained from the Taylor derivation of the Laplacian, and $f_{xy}$ is the difference in $x$ of the difference in $y$.

Question: Can $(\Delta x, \Delta y)$ turn out to be larger than one pixel?

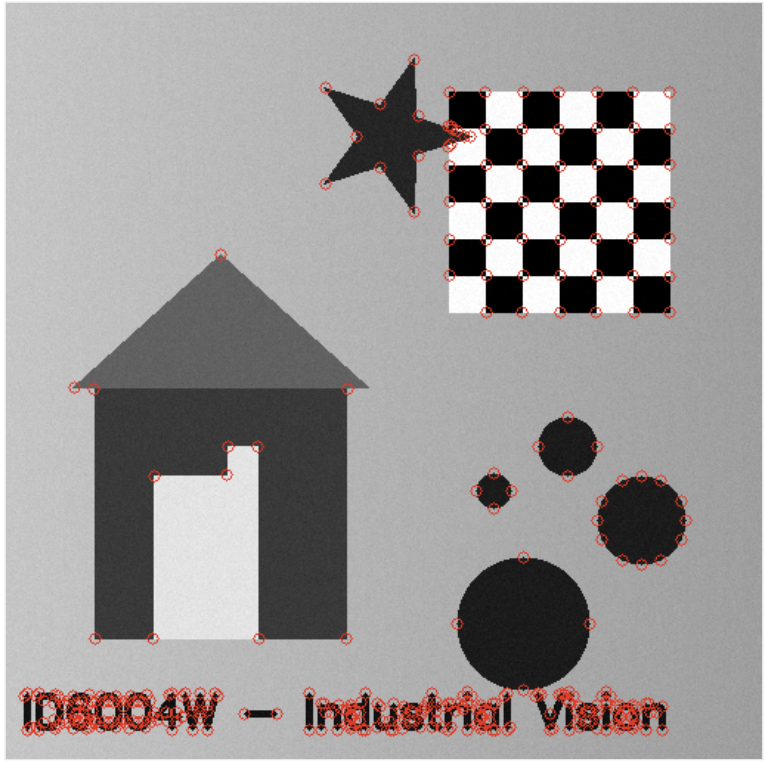
Answer: The expansion is valid only for small $\Delta$. If the solution returns $|\Delta| > 1$, then either the grid maximum was incorrect or the surface is flat.

Implementations therefore either clamp the value or repeat the computation, since this is a single Newton step and may be iterated to convergence.

+ board

# Harris Corner Detector Demo

image Synthetic scene *ideal step edges, exact corners — the textbook case*




κ = 0.050

Harris used 0.04–0.06

threshold T = 1.0% of max M

non-maxima suppression (3×3)

corners response map M

**342 corners flagged**

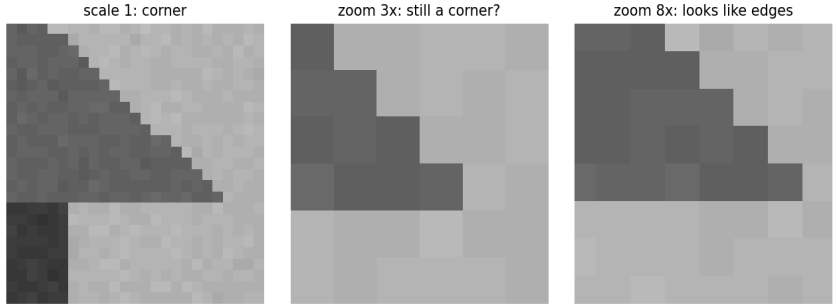
+ board

# Behavior under geometric transformations

| Transformation | Corner location | Corner strength |
|---|---|---|
| Translation | **covariant ✓** | **invariant ✓** |
| Rotation | **covariant ✓** | **invariant ✓** |
| Scaling | **not covariant ✗** | **✗** |

For Rotation: the gradients transform as $\nabla I' = Q^\top \nabla I$, so that $R' = Q^\top R\, Q$. This is a similarity transformation, which leaves the eigenvalues unchanged and rotates the eigenvectors. The corner strength is therefore unaltered and only the orientation changes. This point is worth verifying for yourself.

**Why scaling causes the detector to fail:**




If a corner is magnified, then within any fixed window it degenerates into straight **edges**. The notion of a corner is therefore dependent on scale, and the fixed window is the cause of the difficulty.

+ board

# Behavior under photometric transformations

**An additive shift** $I \to I + a$

Differentiation removes the constant, since $\nabla(I + a) = \nabla I$.

The matrix $R$, its eigenvalues and the response are therefore all **unchanged**, and the detector is fully invariant.

**A change of gain** $I \to a\,I$

Here $\nabla(aI) = a\nabla I$, so that $R \to a^2 R$ and $\lambda_i \to a^2 \lambda_i$, giving

$$M \;\to\; a^4 M$$

The detections themselves survive, since the ordering and the signs are preserved and corners remain corners. A **fixed threshold $T$ does not survive**, however, so the response should be normalized or the threshold expressed relative to the maximum strength in the image.

The Harris detector therefore withstands the two simple photometric changes and the geometric changes of translation and rotation, and its guarantees extend no further.

+ board

# Matching Harris corners between two images

Detection establishes where the corners are. Applications such as stereo, panorama construction and recognition require **correspondence**, that is, a decision as to which corner in one image is which corner in the other.

- The Harris detector provides **no systematic descriptor**. The practical procedure is to take a patch around each corner and to compare the patches.
- If the two views differ by a rotation, one patch is first **de-rotated**, the orientation of each corner being known from the angle of its eigenvectors.
- The patches are compared by **normalized grayscale correlation**. The normalization confers tolerance to differences of illumination, such as those arising from a different day or a different exposure.
- A further check is available at no cost. The **ratio of eigenvalues** $\lambda_1 / \lambda_2$ should agree between matched corners, since rotation preserves it.
- The search may be restricted by assuming a bounded motion and considering only the two or three corners within the corresponding region.

This procedure works, but it is evidently ad hoc. The systematic solution, a descriptor designed for matching that is robust to changes of scale, rotation and illumination, is **SIFT**.

+ board

# Questions

+ board

# Summary

- The **shape** of the gradient distribution classifies a patch, and the decomposition $R = P \Lambda P^\top$ makes this explicit. A flat region has $\lambda_1 \approx \lambda_2 \approx 0$, an edge has $\lambda_1 \gg \lambda_2$, and a corner has both eigenvalues large.
- The **response** $M = \lambda_1 \lambda_2 - \kappa(\lambda_1 + \lambda_2)^2 = \det R - \kappa \operatorname{tr}^2 R$ is approximately zero on a flat region, negative on an edge and positive on a corner, and it requires no eigen-decomposition at run time.
- The pipeline computes the response, applies a threshold, applies **non-maximum suppression** over a $3 \times 3$ neighborhood, and performs a **sub-pixel** Newton step $\Delta = -H^{-1} \nabla f$.
- The detector is invariant to translation, to rotation and to an additive change $I + a$. A change of gain rescales $M$ by $a^4$. A change of **scale defeats the detector**, since a magnified corner appears as a pair of edges.
- Matching is performed by normalized correlation of de-rotated patches. This is serviceable but ad hoc, and it is the reason that descriptors were developed.

**In the next lecture** we address the problem of scale. We study blobs, the scale-normalized Laplacian of Gaussian, and the way in which scale is searched as a third axis alongside the two image coordinates.

+ board

# References


1. Harris, Chris, and Mike Stephens. "A combined corner and edge detector." Alvey vision conference. Vol. 15. No. 50. 1988.
2. Forsyth, David A., and Jean Ponce. Industrial Vision: a modern approach. prentice hall professional technical reference, 2002.
3. Shi, Jianbo. "Good features to track." 1994 Proceedings of IEEE conference on Industrial Vision and pattern recognition. IEEE, 1994.


+ board